

Spatial Columns Fact Sheet

A one-page reference for citing when SQL Server spatial geometry was added to the Viking annotation database.
Prepared for the 3D connectome analysis methods paper.

When were spatial columns added?

The `MosaicShape` and `VolumeShape` spatial `geometry` columns were added to the `Location` table on **19 August 2015**, in commit `ffe9cceb` on the `dev` branch ("Migration to spatial columns in database", author James Anderson). That is roughly **11 years** before the present (June 2026) and about **6 years after** the 2009 PLoS Biology framework paper (Anderson et al.).

The change is recorded in the versioned database build script, [Servers/SQL/DatabaseCreateUpdate/CreateUpdateDatabase.sql](#) (originally `Servers/AnnotationService/SQL/DatabaseCreateUpdate/CreateUpdateDatabase.sql` before later repository reorganization).

What changed (two-step migration)

The migration is applied as two sequential schema versions in the `DBVersion` table:

DBVersion	Date	Change
24	2015-08-19	ALTER TABLE Location ADD MosaicShape geometry and ADD VolumeShape geometry
25	2015-08-19	Populate the new shapes from the legacy scalar columns; drop <code>verticies</code> , <code>X</code> , <code>Y</code> , <code>VolumeX</code> , <code>VolumeY</code> ; re-add <code>X/Y/VolumeX/VolumeY</code> as persisted computed centroids of the shapes

Before this migration, an annotation footprint was stored as scalar columns (`X` , `Y` , `Radius`) plus a `Verticies` vertex list for polygons. After the migration, each `Location` row stores two true geometries:

- `MosaicShape` -- the footprint in the native section/mosaic coordinate space.
- `VolumeShape` -- the same footprint in the registered volume coordinate space.

Circular annotations were converted to `CURVEPOLYGON(CIRCULARSTRING(...))` and point annotations to `POINT(...)` , as seen in the version 25 block of the build script (around lines 2389-2440). The scalar centroid columns are retained as `PERSISTED` computed columns, for example:

```
ALTER TABLE Location ADD X as ISNULL(MosaicShape.STCentroid()).STX,  
ISNULL(MosaicShape.STX,0)) PERSISTED
```

Follow-up spatial commits (same era)

Commit	Date	Summary
<code>ffe9cceb</code>	2015-08-19	Migration to spatial columns in database (DBVersion 24-25)
<code>b579196a</code>	2015-08-21	Further updates for spatial migration

19b70b34	2015-08-26	Add spatial index to the database
64203551	2016-03-24	Spatial queries use mosaic or volume coordinates; synapse-to-lines migration

The same script has since grown to **DBVersion 83**, which improves the spatial bounds stored procedure `SelectSectionLocationsAndLinksInMosaicBounds` .

Methods-ready paragraph (copy/paste)

Annotation geometry in the Viking database was migrated from scalar coordinates (X, Y, radius, and polygon vertex lists) to native SQL Server `geometry` types in August 2015 (database schema versions 24 and 25). Each annotation footprint (`Location`) stores a `MosaicShape` and a `VolumeShape` , representing the footprint in section-mosaic and registered-volume coordinate spaces respectively; the scalar centroid columns (X, Y, `VolumeX`, `VolumeY`) are retained as persisted computed columns derived from those geometries. Spatial indexes (added August 2015) and region/bounds stored procedures support efficient viewport and radius queries during connectome tracing, review, and 3D reconstruction.

How to reproduce these dates

```
# Primary introduction of the spatial columns
git log origin/dev --format="%h %ad %an %s" --date=short \
  -S "MosaicShape geometry" -- "**/CreateUpdateDatabase.sql"

# Inspect the follow-up commits
git show -s --format="%h %ad %an %s" --date=short b579196a 19b70b34 64203551
```